

ULTRON 2.7 Phase 10 Execution Report

Wake-word detection and voice activity gating

Project	ULTRON 2.7
Phase	Phase 10: Wake Word and VAD
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 10 adds wake-word detection and stronger voice activity detection to ULTRON 2.7. ULTRON can now enter an always-listening mode, wait for ULTRON or Hey ULTRON, ignore empty/noisy input, and only send authorized speech segments into STT and the safe runtime.

Wake word and VAD are input gates only. They cannot choose tools, execute commands, skip validation, bypass policy, or bypass audit logging.

Phase 10 Scope

- Add wake-word and VAD provider abstractions.
- Add text/mock wake-word providers and health-checked openWakeWord support.
- Add energy-threshold VAD plus health-checked Silero and WebRTC adapters.
- Add always-listening modes: inactive, waiting_for_wake_word, listening, transcribing, thinking, and speaking.
- Add wake APIs and UI privacy controls.
- Keep push-to-talk and typed input available.
- Add regression tests for wake gating, noisy input, VAD-gated STT, and unchanged high-risk confirmation behavior.

What Was Implemented

Area	File(s)	Result
Wake/VAD runtime	src/ultron27/wake.py	Provider protocols, wake providers, VAD providers, health checks, and wake phrase stripping.
Voice session	src/ultron27/voice.py	Always-listening state, wake snapshots, wake processing, VAD gating, and safe STT handoff.
Local API	src/ultron27/web_server.py	Added wake start, stop, status, and process routes.

Area	File(s)	Result
Configuration	src/ultron27/config.py, ultron.config.example.json	Added wake provider, wake phrases, wake model path, VAD provider, and VAD threshold settings.
Web interface	web/index.html, web/styles.css, web/app.js	Added always-listening toggle, mic privacy status, wake gate status, VAD status, and ignored-input feedback.
Tests	tests/test_pipeline.py	Expanded to 54 tests covering wake word, no-wake ignore, noisy input, VAD-gated STT, wake APIs, config parsing, and safety regression.

Wake/VAD Pipeline

Stage	Purpose	Safety Property
Candidate input	Browser or mock voice submits a candidate speech segment.	No command execution happens at capture time.
VAD	Empty, noisy, or low-energy input is ignored.	Noisy audio does not reach STT or the brain.
Wake word	ULTRON or Hey ULTRON must be present while waiting.	Background speech without wake phrase is ignored.
STT	Authorized speech is transcribed through the existing provider layer.	STT output is still untrusted user input.
Brain/runtime	The command flows through the same typed tool pipeline.	Validation, policy, executor, and audit remain authoritative.

Verification Results

Check	Command	Result
Unit tests	\$env:PYTHONPATH='src'; python -m unittest discover -s tests	54 tests passed.
Compile check	python -m py_compile src/ultron27/config.py src/ultron27/wake.py src/ultron27/voice.py src/ultron27/web_server.py scripts/run_phase10_wake_ui.py	Phase 10 Python modules compiled successfully.
Frontend syntax	bundled node --check web/app.js	Frontend JavaScript parsed successfully.
Dataset evaluation	python scripts/evaluate_dataset.py --split test	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Safety regression	python scripts/evaluate_safety_regression.py	362 cases, tool accuracy 1.0, policy action accuracy 1.0.
Wake API smoke	ThreadingHTTPServer endpoint checks	Wake start, status, process, and stop routes returned expected payloads.

Recommended Next Step

- Install real openWakeWord and Silero/WebRTC dependencies for microphone audio.
- Connect browser microphone chunks or local audio capture to the wake/VAD providers.
- Add streaming partial transcripts after the wake path is stable.